Precalculus B
Assignment
Sec. 4.5 Phase Shifts and Vertical Shifts

Name Alexander
Hour 4 Seat # 6

Direction: Graph each function over a 2 period interval. Label your x and y axis correctly

1. $y = 2\sin 2\left(x - \frac{\pi}{4}\right)$

Amplitude

2

Period

π

Phase Shift

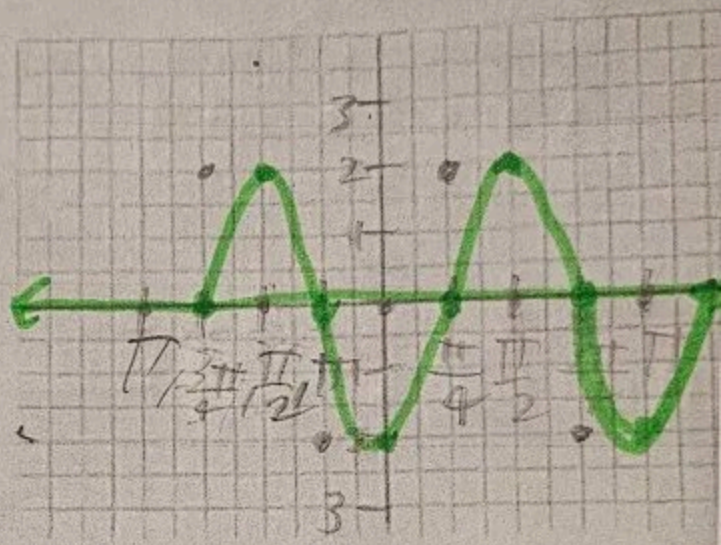
$\frac{\pi}{4}$ right

Guide Graph

$y = 2\sin 2(x)$

Midline:

$y = 0$



2. $y = \cos\left(x + \frac{\pi}{2}\right)$

Amplitude

1

Period

π

Phase Shift

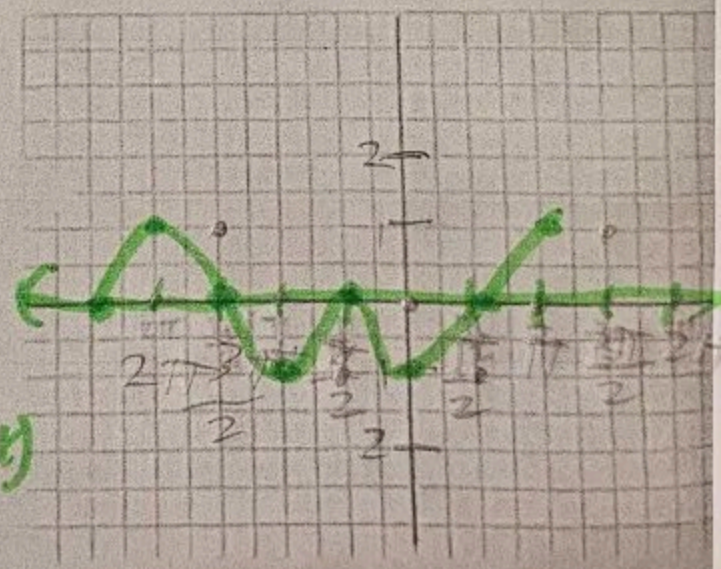
$\frac{\pi}{2}$ left

Guide Graph

$y = \cos(x)$

Midline:

$y = 0$



3. $y = 3\cos 2\left(x + \frac{\pi}{4}\right) + 1$

Amplitude

3

Period

π

Phase Shift

$\pi/4$ left

Vertical Shift

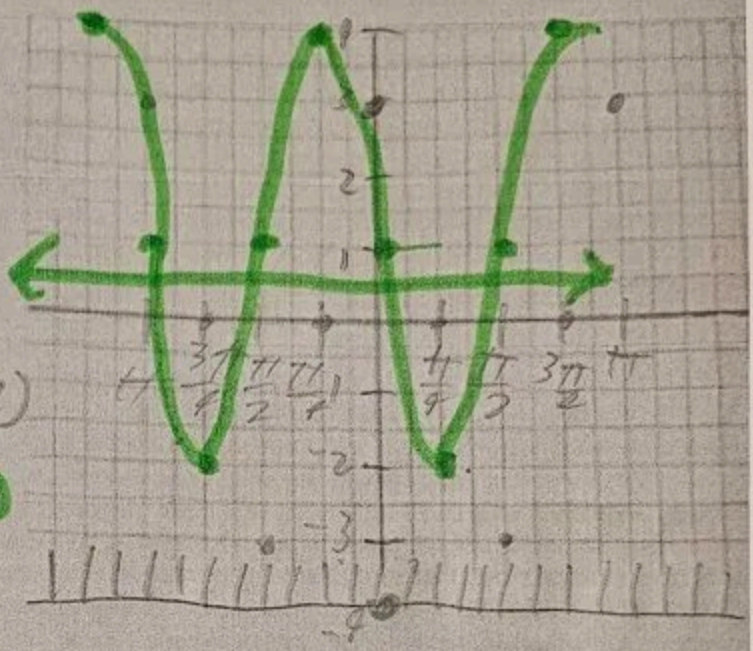
+UP 1 (UP 2)

Guide Graph

$y = 3\cos(x)$

Midline:

$y = 1$



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